

GeoGuide at a Glance

A location-aware travel companion where every AI sentence links to the record it came from, and which says "I don't know" instead of guessing.

The 30-second pitch

"Travellers get generic guides, and AI chatbots that sound sure even when they're wrong. GeoGuide knows where you are and which date you're travelling, briefs you on weather, events, safety and culture for that date, and cites every sentence. Tap any citation to see the exact database row. If the data can't back an answer, it refuses. Every sentence has a receipt."

What it does

- **Arrive:** your location → nearest of 60 cities, weather, nearby places.
- **Briefing:** six cited cards: what to wear, what's on, safety, history, culture, top attractions.
- **Date-Shift:** change the date; events, weather, season, advisories and closures are recomputed. "What changed" lists the differences.
- **Evidence drawer:** tap a citation to see the query and the row, cited field highlighted.
- **Nearby** places and hotels; **Right now:** what fits the next 30–90 min within budget.
- **Ask:** cited answers or a polite refusal, in English, Hindi and Kannada. Read aloud and offline mode.

How it works

1. Browser coordinates → nearest city (not stored).
2. FastAPI reads that city's facts for the date from the read-only SQLite database.
3. For the briefing and chat, ChromaDB finds the 4 closest guide passages by meaning.
4. Gemini writes only from those passages and cites [n] on every sentence.
5. Uncited sentences are dropped; each chip opens its source via /source .

How it avoids making things up

Intent rules	Refuse questions the data can't answer (live fares, traffic, bookings, "ignore your rules") before any AI call.
Relevance gate	Refuse if the best passage scores below 0.30 similarity.
Sentinel	The AI returns a code word when the passages don't cover the question.
Citation check	Every sentence must cite a real passage, or it's removed.

Proof: with grounding off, every question is refused, so the AI never answers from memory. If the AI is down, answers quote the source word for word.

Date-Shift, in three lines

Plain database queries, no AI: events on the date, that day's weather row, the season and peak months, advisories valid at noon, places closed that weekday. Instant and repeatable. An empty date says "Nothing is scheduled" with a 0-rows citation instead of inventing an event.

Numbers to remember

60 cities	3,173 + 900 passages + place facts indexed	68 automated tests
0 AI calls for Date-Shift	2–4 s typical chat answer	1 Sep – 30 Oct 2026 dates the data covers

Stack

React 18, Vite, Tailwind · FastAPI (Python) · SQLite (read-only) · ChromaDB + multilingual MiniLM embeddings · Google Gemini · Sarvam AI (Hindi, Kannada) · Web Speech API · pytest · Vercel + Cloudflare tunnel.

Demo in 60 seconds

1. Arrive: "You are in Bengaluru".
2. Briefing → tap a weather chip → the database row.
3. Tap "Monsoon Music Nights" (17 Oct) → "What changed".
4. Ask "Do I need to remove my shoes at temples?" (cited), then "How much is a cab to the airport right now?" (refused).
5. Right now → select a pick; it glows on the radar.

Quick answers

Why not just ChatGPT?

It answers everything, even when guessing. We only answer from the data, cite every sentence, and refuse otherwise.

What is RAG?

Find the relevant text first, then let the AI write only from it, like an open-book exam.

What if the data is wrong?

Every chip shows its source row, so errors are easy to trace and fix at the source.

Why no login?

Location only picks the city and isn't stored. Nothing kept means nothing to leak.

New city?

Add its rows and rebuild the index. Nothing is hardcoded.

Limits?

Fixed 60-day dataset, no live prices or traffic by design, estimated travel times.

Run it

```
.env/bin/uvicorn backend.main:app --port 8000
cd frontend && npm run dev → localhost:5173
```

First-time setup and the full demo script are in the project guide.